Sharmistha Jat

London, United Kingdom
sharmistha.jat@gmail.com , http://sharmistha.co.in

RESEARCH INTERESTS

Computational Neuroscience, Natural Language Processing and Machine Learning

EDUCATION

PhD, Computer Science, 2015 - Present (Maternity Leave, July 2019 - June 2020)
Indian Institute of Science (IISc), Bangalore, India
Advisor: Prof. Partha Pratim Talukdar and Prof. Tom Mitchell

MTech, Computer Science, 2011-2013
Indian Institute of Technology Madras (IITM), Chennai, India
Advisor: Prof. Balaraman Ravindran

BE, Computer Engineering, 2005-2009
Maharashtra Institute of Technology COE, Pune, India

WORK EXPERIENCE

Xerox Research Centre India (XRCI), Bangalore, Oct 2013- July 2015
Research Engineer, Analytics Lab

Morgan Stanley, Mumbai, Jan 2010 - July 2011
Senior Associate, Technology Division

INTERNSHIP EXPERIENCE

Carnegie Mellon University (CMU), Pittsburgh Sept 2017- April 2018, Sept 2018 - March 2019
Advisor: Prof. Tom Mitchell

PUBLICATIONS

Sharmistha Jat, Partha Talukdar, Tom Mitchell, **Word Salience during Sentence Reading**, Accepted at Human Brain Mapping **(oHBM)**, Seoul, Korea, 2021

Sharmistha Jat, Erika J C Laing, Partha Talukdar, Tom Mitchell, **Spatio-temporal Characteristics of Noun and Verb Processing during Sentence Comprehension in the Brain**, 2020 [under review, preprint available]

Sharmistha Jat, Hao Tang, Partha Talukdar, Tom Mitchell, **Relating Simple Sentence Representations in Deep Neural Networks and Brain**, Accepted at Association for Computational Linguistics **(ACL)**, Florence, 2019

S. Kumar, Sharmistha Jat, Karan Saxena, Partha Talukdar, **Zero-shot Word Sense Disambiguation using Sense Definition Embeddings**, Accepted at Association for Computational Linguistics **(ACL)**, Florence, 2019 [outstanding paper award]

Sharmistha Jat, Tara Pirnia, Tom Mitchell, **Dynamic Time Warping for Brain MEG Analysis**, Accepted at Human Brain Mapping **(oHBM)**, Rome, 2019 [travel stipend award]

Nicole Rafidi, Mariya Toneva, Dan Schwartz, Sharmistha Jat, Tom Mitchell, **MEG Representational Similarity Analysis Implicates Hierarchical Integration in Sentence Processing**, Organization for Human Brain Mapping, **OHBM** 2018, Singapore, June

17-21

Sharmistha Jat, Siddhesh Khandelwal and Partha Talukdar, **Improving Distantly Supervised Relation Extraction using Word and Entity Based Attention**, Automated Knowledge Base Construction Workshop, **NIPS 2017**, Long Beach, USA, Dec 4, 2017

Tushar Nagarajan, Sharmistha and Partha Talukdar, **CANDiS: Coupled & Attention-Driven Neural Distant Supervision**, The First Women and Underrepresented Minorities in NLP Workshop, **WiNLP, ACL 2017**, Vancouver, Canada, July 30, 2017

Lavanya Sita Tekumalla, Sharmistha, **IISCNLP at SemEval-2016 Task 2: Interpretable STS with ILP based Multiple Chunk Aligner**,workshop on Semantic Textual Similarity **(SemEval) NAACL 2016**, San Diego, 707-712

Himanshu Sharad Bhatt, Sandipan Dandapat, Peddamuthu Balaji, Shourya Roy, Sharmistha and Deepali Semwal, **SODA:Service Oriented Domain Adaptation Architecture for Microblog Categorization**, Proceedings of **NAACL 2016**, San Diego,77-81

Anuj Mahajan, Sharmistha, Shourya Roy, **Feature Selection for Short Text Classification using Wavelet Packet Transform**, Proceedings of the 19th Conference on Computational Natural Language Learning, **CoNLL 2015**, Beijing, China, July 30-31, 2015,ISBN - [978-1-941643-77-8], 321-326

Sharmistha, Madhur Amilkanthwar, Shankar Balachandran, **Augmentation of Programs with CUDA Streams**, 10th IEEE International Symposium on Parallel and Distributed Processing with Applications, **ISPA 2012**, Leganes, Madrid, Spain, July 10-13, 2012, ISBN - [978-1-4673-1631-6], 855-856

PATENT GRANTED

Sharmistha Jat, Anuj Mahajan, Shourya Roy, **Method and System for Predicting Requirements of a User for Resources Over a Computer Network**, 2019, United States Patent 10417578

Sharmistha Jat, Koyel Mukherjee, Narayanan U. Edakunni, Pallavi Manohar, **Method and system for recommending one or more vehicles for one or more requesters**, 2018, USPTO Patent Number 9978111

Narayanan U. Edakunni, Sharmistha Jat, **Method and system for determining effect of weather conditions on transportation networks**, 2018, USPTO Patent Number 10062282

Koyel Mukherjee, Ankit Kumar, Pallavi Manohar, Narayanan U. Edakunni, Sharmistha Jat, **Method and system for scheduling vehicles along routes in a transportation system**, 2017, USPTO Patent Number 9746332

COURSES

Machine learning, Computational Methods of Optimisation, Stochastic Methods and Applications, Linear Algebra, Natural Language Understanding, Social Network Analysis, Knowledge Representation and Reasoning, Kernel Methods in Pattern Recognition

Computer Skills

Programming languages (C, C++, C#, Matlab, Java, Python), Frameworks (PyTorch, NumPy, SciPy), Databases (MySQL, MongoDB)

POSITION OF RESPONSIBILITY

Reviewer for ICLR 2021, CoNLL 2019, NIPS 2018, KDD 2018, XRCI Open Innovation Conference 2015

Teaching Assistant for Machine Learning (E0-270), Indian Institute of Science (IISc), January- April 2017

Organized **Girl Student Empowerment workshop** at Department of CDS, IISc, in collaboration with CHETANA Girls organisation.

Tutorial, “Predictive Modeling Using R and Python” at Grace Hopper Conference (GHC), India, 8th December 2016

Teaching Assistant for Social Network Analysis (CS6012), Indian Institute of Technology Madras (IIT Madras), January-April 2013

Member of Charity Committee, Morgan Stanley 2010-2011

HONORS and AWARDS

BrainHub CMU-IISc Fellowship 2017-2018, 2018-2019

AAAI 2019 travel grant, OHBM 2019 travel stipend, NAACL 2016 travel grant

Selected to attend **EMEA workshop for Brain Inspired Computing 2018 and Google NLP summit 2017**

Top 6% in KAGGLE. Member of MALL Lab team which stood 10th among 170 teams for ALLEN-AI Kaggle challenge, demo: http://malllabiisc.github.io/demos/

Best Poster Award at Grace Hopper Celebration India 2016, GHCI is the India’s largest gathering of women technologists with almost 6000 attendees.

Second position winner at Yahoo Hacku, a national level coding competition 2012

Third position winner in Code-Obfuscation coding challenge at Shaastra 2011, a national level competition organised by Indian Institute of Technology Madras (IITM)